

SUSANTA GORAI

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WORK EXPERIENCE

Technical Exposure | CSIR-CMERI

- Worked on innovative projects related to robotics and embedded systems.
- Gained hands-on experience in ROS and Underwater Autonomous and Remote Operated Vehicle.

Robotics Instructor | Asansol Ramakrishna Mission School

- Conducted robotics training sessions for students, covering fundamental concepts.
- Designed hands-on projects to enhance student understanding of robotics and automation.

SKILLS

- **Programming Languages:** Python, C++, C
- **Internet of Things:** Sensor Integration, Wireless Communication, Message Queuing Telemetry Transport
- **Machine Learning & Computer Vision:** Computer Vision, Image Processing, OpenCV, MediaPipe, TensorFlow, PyTorch
- **Robotics & Embedded Systems:** Robot Operating System, Arduino, STM32, ESP32, Teensy, Raspberry Pi, Jetson
- **Web Development:** HTML, CSS, JavaScript, Django
- **Database Management:** MySQL
- **3D Modeling:** SolidWorks
- **Development Tools:** VS Code, Arduino IDE, LIO-SLAM, SLAM Toolbox, Git, Ubuntu

PROJECTS

- **GPS-Less Drone Swarming & 3D Mapping System**
Developed a GPS-denied autonomous swarming drone using 3D LiDAR SLAM for real-time mapping, localization and navigation. Implemented AprilTag-based vision for target detection and autonomous precision landing without GPS.
- **Intelligent ET Sensing & Irrigation System**
Drone-based Intelligent ET sensing system and irrigation water use accounting system for irrigation commands.
- **STM32-Based Flight Controller System PCB**
Designed and developed an STM32-based flight controller PCB for real-time drone control and sensor integration .
- **LiDAR-Based Autonomous Cleaning Robot**
Developed an autonomous cleaning robot using LiDAR, SLAM and ultrasonic sensors for mapping and navigation.
- **IoT-Based Smart Home & Agriculture System**
IoT system with custom PCB integration for voice-controlled home automation and smart agriculture monitoring.

ACHIEVEMENTS & AWARDS

- 🏆 **IRoC-U 2026 Qualifier:** ISRO Robotics Challenge
- 🏆 **Best Applied Technology Project Award:** Science & Engineering Fair 2026, Birla Industrial & Technological Museum
- 🏆 **Winner:** Smart India Hackathon 2024
- 🏆 **Best Innovative Idea Award:** Innovative Hardware Model Making Competition (2024, 2025)
- 🏆 **Best Jury Award:** Innovative Electronics Innovation 8.0 (2023)

EDUCATION

Durgapur Institute of Advanced Technology and Management

Durgapur, Rajbandh, West Bengal | (2023-2027)

Bachelor of Technology in Computer Science Engineering (Ongoing)

CGPA: 7.20

Coursework: Computer Network, Digital-Analog, Data Structure Algorithm, Operating System